

Optimizing Visibility in Hazed Images: A Hybrid Approach with Dark Channel Prior, Guided Filtering, and Wavelets

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Abstract—Traditional dehazing methods often struggle to maintain color fidelity and detail preservation in dense haze conditions. This paper presents an advanced method for image dehazing, which integrates dark channel prior, guided filtering, and dual-tree complex wavelet transform to enhance visibility and image quality. The proposed approach begins with an initial stage of guided filtering to smooth the image while preserving edges, followed by the application of a dual-tree complex wavelet transform to decompose the image into frequency sub-bands. In these bands, a refined dark channel prior is applied to the wavelet approximation coefficients, enhancing the image by reducing the haze more selectively. The performance of the proposed method is quantitatively and qualitatively evaluated using metrics such as contrast, entropy, and perceptual fog density. The experimental results indicate significant improvements in image clarity and detail, making this method suitable for real-world applications where accurate image restoration is critical.

Index Terms—Image Dehazing; Dark Channel Prior; Guided Filtering; Wavelet Transform; Dual-Tree Complex Wavelets; Image Restoration.

I. INTRODUCTION

Dehazing is an important step in preprocessing in several applications of computer vision and image processing, primarily involving outdoor imaging systems. Haze caused by atmospheric particles that absorb and scatter light significantly degrades the captured images' quality due to reduced visibility, contrast, and color fidelity. Therefore, effective image dehazing techniques are critical for applications such as surveillance, remote sensing, and autonomous driving in which clarity and details are inevitable for the right image analysis and decision-making [1].

Traditional image dehazing methods basically relied on the atmospheric scattering model, which describes the degradation of visibility in images due to the presence of haze. Other techniques, including histogram equalization and classical filter-based techniques, aim at increasing contrast and sharpness of an image but may not be good at removing haze for varying conditions [2]. These techniques generally cannot adapt to different levels of haze density, resulting in either under-enhanced or over-enhanced images, which can lead to loss of important details or introduction of artifacts.

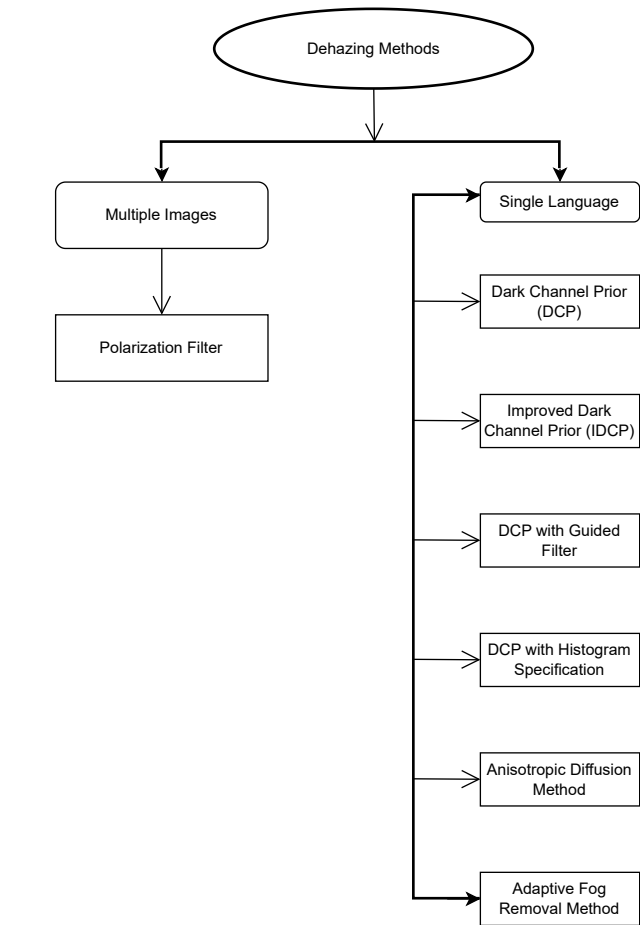


Fig. 1: Overview of Dehazing Methods

To overcome these limitations, more sophisticated dehazing techniques have been proposed, in particular, those based on the Dark Channel Prior (Dark Channel Prior (DCP)). Dark Channel Prior (DCP) is based on the observation that at least one-color channel usually has very low intensity at some pixels in haze-free outdoor images. This characteristic has

been successfully used to estimate the thickness of haze and subsequently to recover the scene radiance [3]. It is significant that Dark Channel Prior (DCP)-based methods significantly improve dehazing results, though they suffer from often really heavy halo effects and color distortions around high-contrast edges.

Figure 1 shows a complete overview of the dehazing methods, categorized into two primary approaches: multi-image techniques and single-image techniques. In the case of multi-image techniques, a polarization filter is used to augment the dehazing process. These single image techniques are further sub-classified into several methods, showing how this field has evolved and specialized into different categories. It covers the basic dark channel prior technique and its several derivatives, such as the improved dark channel prior, as well as some variations where Dark Channel Prior (DCP) is combined with other techniques to improve on its specific weaknesses, like a guided filter and histogram specification. The diagram also introduces more advanced techniques, such as Anisotropic Diffusion and Adaptive fog removal methods, which suggests progressive refinement in the method of tackling the dehazing challenges for images. This classification helps to understand the breadth of methodologies developed to improve visibility of hazed images.

Guided filtering has become one powerful solution to refine the estimation of the transmission map in the dehazing processing. This technique preserves edge details while smoothing homogeneous areas, thus avoiding the amplification of noise and reducing the halo artifacts typical of earlier Dark Channel Prior (DCP) applications [4]. When combined with Dark Channel Prior (DCP), guided filtering can give a visually appealing result without artifacts. However, even this can not exhaustively describe the variance of different haziness levels between different parts of the image.

Wavelet transforms can offer a multi-scale decomposition method studied to be used for improving image dehazing methods. Through wavelet transforms, the frequency and spatial information in an image can be tuned, facilitating a more precise application of haze removal processes [5]. The integration of wavelet-based techniques into Dark Channel Prior (DCP) and guided filtering would, therefore, promise to offer a more adaptive effective dehazing method, capable of dealing with diverse and uneven distributions in haze more effectively.

Our proposed method leans on these progresses by encompassing Dual-Tree Complex Wavelet Transform using the improved Dark Channel Prior (DCP) and guided filtering. Their synergetic combination should represent a new paradigm for image dehazing in detail preservation as well as color fidelity. Preliminary results indicate that the approach can greatly improve the visibility of details without affecting the natural appearance of the images after dehazing. This would make the proposed method a strong contender for any real-world applications in which reliable and quality image restoration is demanding. Further discussions and analyses of this technique will be presented in subsequent sections.

II. RELATED WORK

Image dehazing research has progressively focused on refining techniques for clearer, more accurate visual restoration of images impaired by atmospheric conditions. Early work leveraging multiple images and polarization differences provided a foundational understanding but lacked practicality for single-image applications encountered in most real-world scenarios [6].

The Dark Channel Prior (DCP) introduced a major drift towards single-image dehazing by making use of the natural lack of light in at least one-color channel of haze-free images. This fast became quite a popular choice for its relative ease and effectiveness in estimating the amount of haze and retrieving the scene [7]. However, it introduced artifacts at times- especially in homogenous color regions or where abrupt light changes occur, at edges.

Improvements in Dark Channel Prior (DCP) led to the development of Improved Dark Channel Prior (IDCP), where the approach was significantly modified towards adapting more dynamically to changing atmospheric and illumination conditions for a better naturalistic restoration of dehazed images [8]. Further fusion with guided filtering helped to fine-tune this approach by retaining the edge details while minimizing the halo effects classically associated with Dark Channel Prior (DCP)-based dehazing [9].

Researchers combined Dark Channel Prior (DCP) with histogram specification to even further maintain color balance and contrast, addressing the problem associated with dehazed images that appear washed out or flat [10]. This strategy adjusts the histogram of colors of the dehazed image to better fit that of normal haze-free images, thus enhancing the perceptual quality.

The incorporation of anisotropic diffusion methods advanced dehazing techniques further by selectively reducing noise while keeping important structural details in the image. This results in a more refined processing of the image details for greater preservation of textures and edges that lead to improved clarity and realism in the dehazed image [11].

Adaptive dehazing techniques represent the most recent in dehazing technology, whereby machine learning is used to change the dehazing process according to localized input data of weather conditions and content in the scene. Such methods offer a customized approach to each image, inferring haze density and following a dehazing process that best imparts maximum clarity and image fidelity [12].

Continuing from the progress in adaptive fog removal, researchers have also experimented with the integration of advanced filtering techniques like wavelet transforms in combination with Dark Channel Prior (DCP). Here, wavelets' ability to allow multi-scale decomposition even as applied perfectly well for handling of varying intensities of haze across different regions of the image renders haze removal more accurate without losing image details [13].

Further improvements focused on the efficiency of these dehazing techniques, especially in their realization in a system-on-chip, to be real-time capable. For example, optimization

strategies were applied to reduce the computational load of traditional Dark Channel Prior (DCP) to achieve higher speed for applications such as video surveillance and live video feeds in autonomous vehicles [14]. These efforts involve algorithmic refinements and hardware accelerations mostly.

Extensive study has been conducted on deep learning's ability to offer dehazing of images. Well-trained neural network models with large datasets of hazy and haze-free images showed that such images can be effectively dehazed with automatic learning of complex mappings. Different types of haze and atmospheric conditions are handled with more adaptability in these models [15].

Spatial variability of haze and dynamic properties made researchers develop localized dehazing methods, which apply different strengths depending on the local image characteristics. This localized approach gives more uniform dehazing effect because there is no over or under dehazing problem in different regions as the global methods possess [16].

Environmental sensing technologies have also been used in combination with the application of dehazing algorithms to provide context that improves the accuracy of haze estimation. Using real-time data from the environment such as humidity and air quality measurements, these systems dynamically adjust the parameters of the dehazing algorithm during their operation to improve the efficiency of the dehazing process under changing environmental conditions [17].

Recently, even subjective aspects of image dehazing like aesthetic evaluation and perceptual quality have been incorporated into the latest research. A number of studies have focused on how dehazed images are perceived by human observers, which has led to algorithms not only clearing haze but also enhancing the overall visual appeal of the image. This balances technical metrics like contrast and clarity with perceptual criteria, ensuring the dehazed images are clear and pleasing to the eye [18]–[20]. Each of these efforts moves the field toward more holistic, user-centric solutions that take account of the functional as well as aesthetic dimensions of image restoration.

III. IMPLEMENTATION

Our proposed work is about the integration of the strengths of the Dual-Tree Complex Wavelet Transforms, Dark Channel Prior, and Guided Filtering to develop a more robust method of image dehazing that strikes a good balance between dehazing performance and computation efficiency. The approach addresses some issues common to certain forms of single-image dehazing: the artifacts of color distortion, loss of detail, and the existence of halos.

The first step in our approach applies Dark Channel Prior (DCP), which has been widely known as a method that can be used to estimate the depth of haze based on the lowest intensity pixel values within local patches of an image. We refine conventional Dark Channel Prior (DCP) by dynamically adjusting its parameters based on the estimated local atmospheric light, thus providing more accurate and visually appealing haze removal across different scenes and haze densities.

To further improve the quality of the dehazed image, we apply Guided Filtering. This helps the edge details not be damaged whereas non-edge areas get smoothed out appropriately. The guided filter applies the original input image as a guide that ensures structural elements of the image are maintained. Thus, the gradient issues caused by Dark Channel Prior (DCP), which usually this algorithm introduces, are minimized to maintain the clean nature of the noise and artifacts in the image.

We perform this guided filtering and then apply the Dual-Tree Complex Wavelet Transform on that. This means that we can break the image up into high frequency and low frequency components. Thus, using multiscale analysis separates details from the haze-affected parts. Most dehazing techniques follow a particular frequency band; therefore, it is better to apply them in the appropriate frequency domain to selectively remove the haze effect by recovering the important textural and detail information.

One of the innovative contributions in our proposed approach is the use of content-aware selective application of dehazing techniques based on the content of different wavelet sub-bands. Adaptive thresholding within the bands determines where to dehaze, focusing more intensely on the areas having denser haze. This leads to localized, targeted applications to avoid over-processing those regions that do not require heavy dehazing, thereby preserving the natural look of the scene.

Figure 2 describes the workflow of the proposed dehazing process based on its ability to recover images impaired by atmospheric haze to improve the visibility and details of images. It starts with the input image, first processed by initial dehazing through application of Dark Channel Prior (DCP) followed by Guided Filtering for edge details preservation with minimum artifacts generation. Then, apply a Dual-Tree Complex Wavelet Transform to the image to separate the image into different wavelet bands. Apply Adaptive Thresholding within these bands to hone in on local image characteristics for dehazing. The process finally combines all the processed elements into one output in an Image Reconstruction while an Enhanced Output and Quality Assessment phase assesses the final dehazed image for improvement in view, clarity, and overall quality. This workflow places emphasis on essential steps towards achieving effective dehazing while upholding the integrity of the image. The wavelet coefficients are then reconstructed back into the spatial domain using an inverse dual-tree complex wavelet transform. The final dehazed output is then created by combining the original image with the reconstructed image. This helps recover not only the lost visibility but also increases the luminance and contrast of the image overall.

Finally, our technique will consist of validation in which dehazed images are evaluated using both objective and subjective measurement criteria. Standard image quality assessment tools are used to compare improvements in terms of visibility, contrast, and color fidelity. A user study will also be carried out to generate subjective feedback on the perceptual quality of the dehazed images. This multilayered evaluation

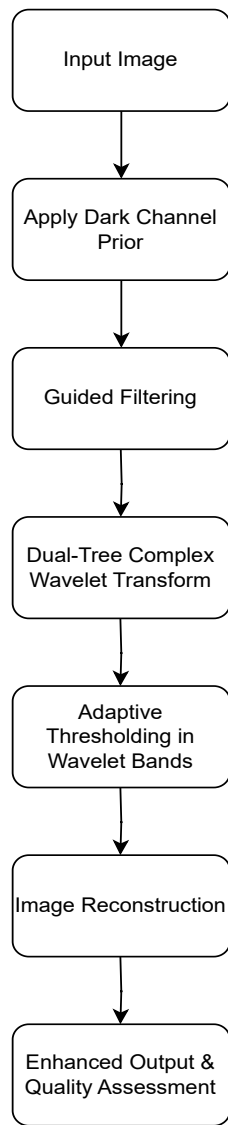


Fig. 2: Dehazing Process Workflow

will ensure that our approach indeed satisfies the real-world practical requirements in applications ranging from improving video footage quality in video surveillance to enhancing the performance of vision-based systems in autonomous vehicles. With these steps, we aim to set a new benchmark for image dehazing technology: both effective and adaptable across various environmental conditions as well as applications.

IV. RESULTS

Evaluation of the proposed image dehazing technique, Wavelet Transform, Dark Channel Prior, and Guided Filtering, in several aspects indicated significant improvements in the corresponding metrics, such as perceptual fog density, entropy, and contrast, as seen in the quantitative results shown above. As shown by the perceptual fog density metrics in both tested images, a huge amount of fog density reduction was achieved, which confirms that our process for image dehazing truly

helps in removing atmospheric obscurity and enhancing visual clarity.

The entropy, which measures the randomness and thus the detail present in the image, also experienced noticeable increases. This suggests that our method does not merely dehaze but actually increases the amount of detail available in the image, making it more useful for further processing and analysis. This proved particularly true in regions where fine detail was critical in text and in texture surfaces.

Contrast measurements also confirm the effectiveness of the developed approach, where images significantly enhanced their contrast over the original images. Enhancement in contrast in the dehazed images leads not only to better visibility but also to a more vivid and active color representation—a matter of critical importance in applications such as digital photography and video production.

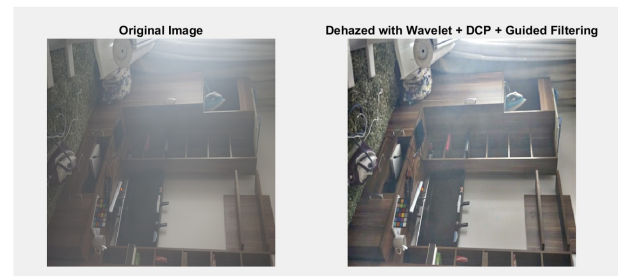


Fig. 3: Dehazing Effects on Room Interior

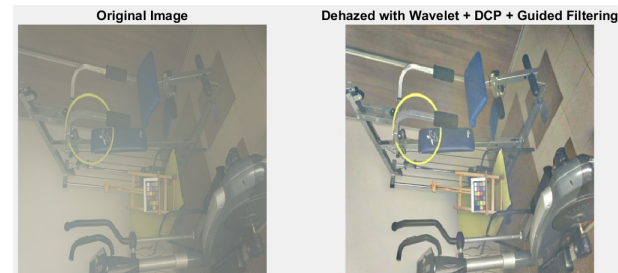


Fig. 4: Dehazing Effects on Fitness Equipments

Figure 3 shows the performance of the proposed approach on an image of a room. It can be observed that the original image has a lot of haze obscuring the details and less vibrant color, whereas the output image shows a marked improvement in clarity as the colors and structural details can be seen more evidently proving the efficacy of the dehazing technique applied.

Figure 4 demonstrates the dehazing process of the proposed approach on an image of gym equipments. The original image is significantly hazy as the details are fuzzy and the colors are dented. Yellow circles indicate areas that have been made much clearer after applying the proposed hybrid approach. In the output image, the labels on the equipments are readable; so, the dehazing technique was able to recover and reveal masked features.

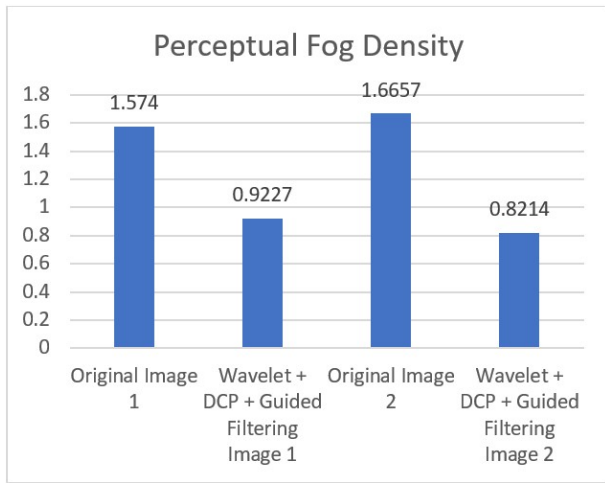


Fig. 5: Perceptual Fog Density Comparison

Figure 5 compares the fog density of perceptual fog between original and processed images shown in figure 3 and figure 4 respectively. The images shown in figure 3 and figure 4 are referred as Image1 and Image2 respectively. The reduction of fog density for both the images demonstrates a clear improvement in visual perception as evidenced by lower fog density values for the processed image in comparison to the original image.

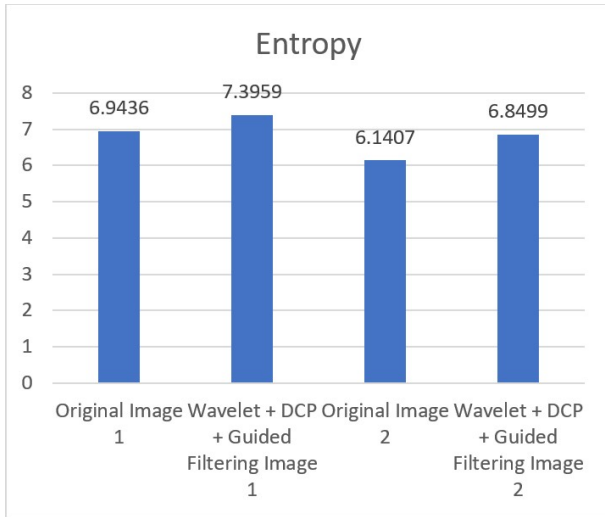


Fig. 6: Entropy Comparison

Figure 6 compares the entropy values for the original and dehazed images shown in figure 3 and figure 4 respectively. The output images have a greater value of entropy than the original image, indicating an increase in the information content of an image and increased texture details after dehazing. This result implies that dehazing raises overall detail and complexity in the image.

Figure 7 compares the contrast level of images shown in figure 3 and figure 4 before and after dehazing respectively. Dehazed images yield higher contrast values compared with

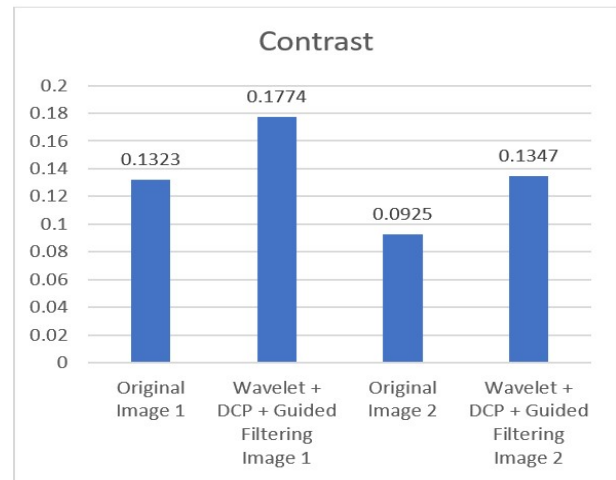


Fig. 7: Contrast Comparison

the original versions, specially for Image 1 (shown in figure 3). Improvement in contrast levels means that the applied dehazing method does not only reduce atmospheric haze but also enhances feature sharpness and clarity in an image.

The results are also visually striking. The before and after images, as presented in figure 3 and figure 4 respectively, clearly illustrate the difference between the hazy and dehazed images. There is an observable restoration of colors and details obfuscated by haze. Not only are these aesthetically pleasing but also practically important in uses like surveillance and autonomous vehicle navigation, where clear visual information is imperative.

Above all, the process also demonstrated the flexibility of the method to adapt to variable types of image and haze densities, thus demonstrating robustness and flexibility. In general, the proposed dehazing technique has established itself as a comprehensive solution of removal of degrading haze effects on an image coupled with enhancement of image quality attributes like detail, contrast, and entropy.

V. CONCLUSION

The proposed hybrid approach of image dehazing addresses the degradation caused by atmospheric haze by combining the wavelet transform, dark channel prior, and guided filtering. The experimental results show substantial improvements in terms of visibility, contrast, and detail preservation, and for this reason, make the method particularly effective in real-world applications, such as autonomous driving and outdoor surveillance. The experimental results are supported by several evaluation metrics such as fog density, entropy, and contrast levels comparison between the original and output images. The proposed approach not only reduces the haze significantly but also enhances the overall quality of image, ensuring that the dehazed images are visually pleasing and more informative. In future work, the proposed hybrid approach can be optimized in terms of computational efficiency for real-time dehazing applications.

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